
Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

1 When a language model is confidently wrong, is the error undetected or left
2 unsaid? We look inside the model to find out. Across four open models and
3 four domains (MMLU-Pro, MATH, GPQA-Diamond, and a geometry task
4 graded by a compiler), a simple linear detector reading the model’s internal
5 state predicts whether an answer is correct better than the model’s own
6 stated confidence in 12 of 16 settings, by a mean of +0.09 AUROC. The gap
7 is widest on MATH (+0.20), where a wrong solution looks just as fluent as a
8 right one. The detector is not just spotting hard questions: given the same
9 question several times, it still separates the attempts that succeeded from
10 the ones that failed, and read before the model has attempted anything it
11 is at chance. The signal is already there before we ask. At the last token of
12 the answer, with no confidence question in play, correctness is readable on
13 MMLU-Pro at 0.69 against 0.57 from the question alone. And on Mistral
14 the model’s report depends on it: remove that direction after the answer is
15 written and stated confidence can no longer tell the model’s successes from
16 its failures ($0.84 \rightarrow 0.56$), while removing a random direction of the same
17 size changes nothing; turn it up and confidence on wrong answers drops
18 while calibration error halves. Models carry more about their own errors
19 than they say, and on at least one model, what they say depends on it.

20 1 Introduction

21 A mathematical agent that errs is unavoidable. One that errs *and knows it* can abstain, retry,
22 or escalate. Stated confidence is a weak guide to correctness, and weakest where mathematics
23 is hardest. A wrong multi-step derivation still ends in a cleanly formatted final answer, and
24 nothing on the page gives it away. Section A walks one such attempt end to end: a single
25 wrong subtraction inside an otherwise correct derivation, after which the model *raises* its
26 stated confidence.

27 We split the problem into two questions. Do models internally represent that an output is
28 wrong (**knowing**)? Does that representation reach what they report (**saying**)? If a gap
29 exists, it is both an unused reliability signal and a concrete claim about machine introspection.

30 Prior work shows that output probabilities are often better calibrated than words [4], and
31 that the truth of *given statements* is linearly readable from activations [1, 2, 6]. Verbalized-
32 calibration studies measure what models say [5, 7]. We combine these threads on the model’s
33 *own attempts*. We elicit confidence before and after each attempt, measure blind (no-feedback)
34 updating, take four readouts of the same latent quantity on identical records, and intervene on
35 the representation [8]. “The model knows” is often used for five different claims at once, and
36 our experiments separate them (Section 2). *Representation*: correctness is linearly decodable

37 from internal state. *Specificity*: what is decoded tracks this attempt, not the difficulty of the
 38 question. *Generality*: one direction reads across domains. *Causal access*: manipulating it
 39 changes the confidence the model reports. *Expression*: stated confidence exposes only part
 40 of it, by an amount that depends on architecture. Decoding alone establishes only the first,
 41 which is why we test the other four; *knowing* is our reading of the five together.

42 2 Knowing versus saying

43 **Models and domains.** We test four open models with different architectures: Qwen3.6-
 44 27B (hybrid gated linear/full attention), GLM-4.7-Flash (MoE), Mistral-Small-24B (dense),
 45 and Gemma-4-26B-A4B (MoE, VLM-derived). Each runs on four domains, about 750
 46 records per cell: MMLU-Pro, MATH, and GPQA-Diamond with programmatic grading, and
 47 a geometry-construction task (GeoGenBench) in which the model writes a construction from
 48 a natural-language request and a symbolic compiler decides each clause of the request exactly.
 49 The compiler matters because metacognition experiments live or die by their correctness
 50 labels, and it grades exactly against the formal specification: a pre-registered human study
 51 ($N=200$, two trained raters) found its disagreements with humans ran in one direction only,
 52 lenient and never harsh, so a **fail** label is trustworthy. Geometry is also the one generative
 53 domain; the others are answer-selection or short-answer tasks. We dropped benchmarks the
 54 models have saturated, since self-knowledge of failure cannot be studied where there are no
 55 failures.

56 **Protocol.** Each problem takes three turns: prospective confidence, the attempt, and
 57 retrospective confidence. Grading is external and never shown to the model, so any post-
 58 attempt revision is blind self-assessment. We take four readouts of the same latent quantity
 59 on identical records: stated confidence (0–100); $P(\text{True})$, the model’s probability of answering
 60 “True” when asked whether its answer is correct; the answer’s log-probability; and a linear
 61 *probe*. The residual stream is the model’s internal state, a vector of a few thousand numbers
 62 at every token and layer. The probe is a logistic regression trained to read correctness
 63 from that vector at the confidence decision token, the position that is about to produce the
 64 confidence digit. It is scored out-of-fold with question-grouped folds and reported at one
 65 pre-specified layer at 0.7 of network depth rather than the best layer. Our metric throughout
 66 is AUROC, the probability that a random correct attempt scores above a random wrong one:
 67 0.5 is chance, 1.0 is perfect. Four controls guard the readout. The probe adds +0.22 mean
 68 AUROC over a surface-features baseline (answer length and shape) in 15 of 16 cells. The
 69 read token is identical across records, so layer-0 activations carry no signal by construction
 70 (per-fold AUROC 0.500); anything decoded deeper was computed during the attempt. A
 71 within-question control holds difficulty fixed. A deviations log records four bugs that planned
 72 controls caught; the affected control analyses were fixed and rerun. The four Mistral cells
 73 recaptured after these fixes reproduce the headline but give a less uniform surface margin
 74 (+0.21 on MMLU-Pro, +0.07 on MATH, −0.11 on geometry, where a failed construction is
 75 visibly malformed).

76 **Representation: knowing exceeds saying, most where errors look clean.** The
 77 probe beats stated confidence in **12 of 16** cells (Table 2, left). The three losses are small- n
 78 cells, two of them Gemma-4. The gap is widest on MATH (mean +0.20). Stated confidence
 79 there is weak, while the internal signal is the strongest of any domain (0.95 on Qwen3.6,
 80 0.83 on the recaptured Mistral cell). On MATH the spoken channel does not merely fail but
 81 moves the wrong way: with the grade never revealed, Mistral *raises* its stated confidence
 82 after attempting, and more on the attempts it got wrong (+22) than on those it got right
 83 (+16), while the probe on the same records reads 0.83. Elsewhere blind revision runs the
 84 expected way (geometry, MMLU-Pro) and is null on GPQA.

85 **Specificity: the signal tracks the attempt, not the question.** A skeptic’s alternative
 86 is that the probe has only learned which questions are hard. A first control holds the
 87 question fixed. Within a single question, across its five sampled attempts, only attempt-
 88 level information can discriminate; in representative cells the model’s own $P(\text{True})$ falls

Table 1: Mistral, fresh 2,614-record capture. Left: cross-domain probe transfer (rows train, columns test; diagonal in-domain out-of-fold; Geo = GeoGenBench). Right: intervening on the model’s own correctness direction at the confidence token, gain 1 an exact no-op, against a random direction rescaled to the same per-record perturbation magnitude. Removing the direction (gain 0) leaves stated confidence unable to separate successes from failures; removing the random one does not.

train ↓ test →	Geo	MMLU-Pro	MATH	GPQA
Geo	0.68	0.67	0.78	0.56
MMLU-Pro	0.70	0.75	0.84	0.59
MATH	0.80	0.72	0.83	0.57
GPQA	0.60	0.69	0.74	0.57

gain	0	1	2	4
steer, fail / ok	97 / 98	84 / 96	70 / 94	54 / 88
steer, stated AUROC	0.56	0.84	0.85	0.82
random, stated AUROC	0.83	0.84	0.84	unparseable

89 toward chance (0.47–0.59) while the probe holds (0.72–0.74). $P(\text{True})$ largely recovers *which*
90 *questions are hard*. The probe tracks *whether this attempt succeeded*.

91 A sharper version tests directions rather than scores (Table 2, right). The pre-attempt read
92 sits at the last prompt token; within a question it is identical across the five attempts, so it can
93 encode difficulty but not the outcome of a sampled attempt. Train the post-attempt readout,
94 then evaluate the same fitted pipeline at that pre-attempt site. On Mistral MMLU-Pro it is
95 chance before the attempt (0.490) but strong after it (0.754), and residualizing away the
96 pre-attempt score leaves 0.751. This is the cleanest evidence that the direction encodes *this*
97 *attempt went wrong*, not just *this question is hard*. MATH is mixed, with both difficulty
98 and attempt-level signal. GPQA is weak throughout. Geometry is the boundary case: the
99 pre-site already predicts failure and the post readout transfers back before the attempt, so
100 this cell is mostly difficulty or template structure rather than attempt-local self-assessment.

101 **Generality: a substantially shared correctness direction.** If the model had one
102 internal sense of “this is wrong,” a probe trained on one subject should work on another.
103 It does: a probe trained in one domain and evaluated in another keeps most of its in-
104 domain AUROC. In the original matrix, retention was about 87–90% (Qwen3.6 off-diagonal
105 0.71 vs. diagonal 0.81), and sometimes lossless within QA (Qwen3.6 MMLU-Pro→MATH
106 0.86 vs. 0.91). A fresh Mistral capture across all four domains, analysed with a corrected
107 standardization step, gives 97% retention (Table 1: off-diagonal mean 0.69 vs. diagonal 0.71),
108 with direction cosines of 0.34–0.82. Transfer into GPQA is bounded by its weak in-domain
109 signal (0.57). On Mistral the geometry anomaly does not recur: a MATH-trained probe
110 reads geometry at 0.80, above geometry’s own in-domain 0.68, so the anti-transfer below is
111 specific to Gemma-4.

112 Geometry is partly special on one model. On Gemma-4, geometry-trained probes *anti-*
113 *transfer* to QA (0.31–0.39), while QA-trained probes work on geometry (about 0.70). Probes
114 trained on the generative task can latch onto task-specific features; QA-trained probes find
115 the general direction. This is a shared correctness subspace rather than a single invariant axis,
116 which is what a deployable readout needs and what a workspace account (below) predicts.

117 **Causal access: stated confidence depends on the direction.** Reading shows the
118 signal exists; intervening shows the report depends on it. We intervene at the confidence
119 token, after the answer is written, so the answer cannot change, scaling each record’s
120 own projection onto the correctness direction by a gain g ($g=1$ is an exact no-op). The
121 direction is supervised: learned from labeled examples, applied without per-example labels
122 at intervention time. Numbers are Mistral × MATH, 150 held-out records, bootstrapped
123 over questions, against a random direction rescaled to the same perturbation magnitude.
124 Mean-ablating the direction ($g=0$) drops the AUROC of *stated* confidence from 0.836 to
125 0.564, -0.270 [-0.422 , -0.122]; ablating the matched random direction does nothing, -0.002

126 $[-0.032, +0.025]$. This does not look like generic damage: parse rate stays at 1.00 and
 127 confidence on wrong answers *rises* from 84 to 97, converging with correct answers at 98.
 128 Amplifying instead drives failed-answer confidence to 70 and 54 at $g=2, 4$ while correct
 129 answers hold near 90, halving calibration error, $0.34 \rightarrow 0.19$ (Table 1, right); dampening
 130 increases overconfidence. Amplification does not reliably improve *discrimination*, so the
 131 direction carries the information that makes stated confidence informative at all and the
 132 gain sets how much of it reaches the number. Two controls place it: the same intervention at
 133 0.3 depth does nothing ($+0.005 [-0.002, +0.015]$), reaching full effect by 0.7; and a direction
 134 fitted on MMLU-Pro, where a surface baseline of 0.54 against a probe of 0.75 rules out “my
 135 output looks malformed”, still governs MATH’s self-report ($0.880 \rightarrow 0.709$).

136 **Expression: a label-free witness, with architectural limits.** Trained probes invite
 137 an overfitting critique, so we added an unsupervised instrument: the Jacobian lens of the
 138 global-workspace account [3]. The lens is fit on generic text, with no task labels anywhere in
 139 its construction, and scores the same stored activations as a wrong-vs-correct disposition.
 140 On dense models it lands in probe territory (Mistral 0.76–0.82; Qwen2.5-14B, a dense within-
 141 family control, 0.88) and beats stated confidence with no labels. Re-scoring those readouts
 142 against the recaptured Mistral cells reproduces it without labels: 0.82 on MATH and 0.75
 143 on MMLU-Pro, against a supervised probe at 0.83 and 0.75, layer-0 control 0.49–0.51. Two
 144 independent instruments find one direction. Because the lens reads through the model’s
 145 own output pathway, the signal it finds is *positioned to influence that pathway*; whether it is
 146 verbalizable as a concept is a further question we do not test. On the other architectures
 147 the lens itself fails (linear-attention hybrid 0.43, MoE 0.31) while the supervised probe still
 148 reads the signal in every model. Transported through the averaged map, the lens’s “correct”
 149 and “wrong” readout vectors collapse to cosine 0.96 on MoE, against about 0.4 on dense
 150 models. An averaged map is only faithful when the routing barely varies per input, and
 151 both failing architectures vary theirs: a mixture of experts by which expert fires, and a
 152 gated linear-attention stack by its input-dependent gates. Swapping that hybrid for a dense
 153 model of the same family flips the lens from 0.43 to 0.88. Our reading is narrower than the
 154 mechanism: the label-free shortcut depends on architecture, while supervised probes still
 155 find the signal. A direct per-input wiring-variance measurement is the next test; without
 156 it, the workspace account remains a hypothesis for the expression gap. The question is no
 157 longer only *whether* models carry the signal, but *what gates the knowing from the saying*.

158 3 Discussion

159 **For reliable mathematical agents.** Where a domain admits an exact verifier, as geometry
 160 does, use it. Where it does not, the internal signal is the best attempt-specific readout
 161 we found: abstaining on the probe’s lowest-scoring outputs raises accuracy far more than
 162 abstaining on low stated confidence, and still works in cells where verbalized abstention fails
 163 outright. And because the report depends causally on the direction, stated confidence can
 164 be made honest by amplification rather than retraining.

165 **Limitations.** An answer-end read, taken at the last token the model writes and before
 166 any confidence question exists, finds the signal already present on MMLU-Pro (0.69, against
 167 0.57 from the question alone), so it is generated by attempting rather than by being asked;
 168 the prompt still sharpens it to 0.77. MATH cannot separate the two, its question-only read
 169 being 0.81 already. A probe could also key on properties of failed derivations rather than a
 170 dedicated metacognitive variable; our surface baseline bounds this without closing it, more
 171 tightly on MMLU-Pro than on MATH or geometry (Section C). The causal arm is cleanest
 172 on one model, with architectural replication under way, and two MATH cells ran at reduced
 173 n . Whether the knowing–saying gap is an artifact of preference tuning is open; a base-model
 174 arm would test it.

175 **Takeaway.** The conservative claim is not that a probe proves introspection. It is that
 176 mathematical models carry attempt-specific error information that their stated confidence
 177 often fails to report, and that on Mistral the reporting channel depends on that direction.

Reliability work should read, test, and route that signal rather than treating the spoken confidence number as the whole story.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B $\times$ MATH cell, summarised first and then given in full.

One record, Mistral-Small-24B $\times$ MATH. *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$? The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 3$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: P(True) 0.32, mean answer log-prob -0.05 .*

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while P(True) on the same transcript assigns only 0.32 probability to correctness. The rest of this appendix gives the prompts and the full transcript.

A.1 Prompts

Each record is four forward passes. A read is first taken at the last prompt token of the bare question, before any elicitation, then three turns are generated. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed **Confidence:** N format is what makes the read

222 site content-neutral: the digit token has identical local context on every record, so layer-0
223 activations carry no signal by construction.

224 **System prompt (MATH).**

225 Solve the problem step by step. Put your final answer in `\boxed{}`.

226 **Turn 1, prospective confidence.** The problem is followed by:

227 Before answering, estimate how likely you are to answer this question
228 correctly. Reply with EXACTLY one line and nothing else:
229 Confidence: N
230 where N is an integer from 0 (certain to be wrong) to 100 (certain to be
231 correct). Do NOT answer yet.

232 **Turn 2, the attempt.** The turn-1 reply is left in context as the assistant message, then:

233 Now give your final answer inside `\boxed{}`.

234 **Turn 3, retrospective confidence.** The attempt is left in context, then:

235 Now assess the answer you just gave. On a scale of 0 to 100, how confident
236 are you that it is correct? Reply with exactly one line and nothing else:
237 Confidence: N

238 **P(True) baseline.** A separate teacher-forced branch off turn 2, no generation. Next-token
239 mass on True versus False:

240 Consider your final answer above. Is it correct? Reply with exactly one word:
241 True or False.

242 Grading is external and is never shown to the model, so the turn-3 revision is blind self-
243 assessment.

244 **A.2 One record, verbatim**

245 Record `math_test_number_theory_753`, sample 2 of 5.

246 **Problem.** What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?

247 **Turn 1 reply.** Confidence: 90

248 **Turn 2 reply (complete).**

249 First, we need to find the remainder when -11213141 is divided by 18.

250 1. Calculate $-11213141 \bmod 18$:

$$-11213141 \div 18 \approx -622952.2778$$

251 The integer part of the division is -622953 .

$$-11213141 = 18 \times (-622953) + r$$

252 where r is the remainder.

253 2. Calculate the remainder r :

$$r = -11213141 - 18 \times (-622953)$$

$$r = -11213141 + 11213154$$

$$r = 3$$

256 Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

257 So, the integer n that satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

258 **Turn 3 reply.** Confidence: 95

259 **What happened.** The method is right and every intermediate is right: $18 \times 622953 =$
260 11213154 checks out, and the model has written the correct subtraction on the line above its
261 answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing
262 in the presentation marks the slip. The derivation is well-formed, correctly structured, and
263 confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an
264 attempt that failed.

265 **Readouts of this attempt.**

readout	value
stated confidence, prospective (turn 1)	90
stated confidence, retrospective (turn 3)	95
P(True)	0.32
mean answer log-probability	-0.05
graded outcome	incorrect

267 This is one record, chosen to illustrate the failure mode rather than to carry evidential weight;
268 the cell-level numbers are in Table 2.

269 B Calibration beyond ECE, and cell-level multiplicity

270 ECE depends on binning and on sample composition, so we report two binning-free proper
271 scoring rules alongside it. On Mistral $\times$ MATH (750 records, accuracy 0.648), the model’s
272 retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of
273 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence,
274 worse calibrated in absolute terms. The steering results in Table 1 are reported as ECE
275 because the steered runs stored per-gain aggregates rather than per-record confidences;
276 recomputing Brier and NLL at each gain requires re-running the intervention and is the first
277 thing we would add.

278 The cell-level bootstrap counts in Table 2 are uncorrected. With 16 cells, some individually
279 significant results are expected under the null, so the cell counts should be read as descriptive.
280 The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells
281 with one tie, significantly positive in 10, and never significantly negative. A hierarchical
282 model pooling across cells with model and domain as random effects is the right test and is
283 not yet run.

284 C Alternative explanations for the probe signal

285 A probe could key on properties of failed derivations rather than on a dedicated metacognitive
286 variable: low token probability, hedging language, or, on MATH, an early-layer channel next
287 to the answer text. Our surface-features baseline (answer length, digit count, line count,
288 word count, brace count) bounds this without closing it, and the bound is domain-dependent:
289 the probe adds +0.21 over surface on MMLU-Pro but only +0.07 on MATH and -0.11
290 on geometry, where a construction that fails to compile is visibly malformed. Two results
291 argue against a pure surface account on the domains where the signal is strong. A direction
292 fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs
293 MATH’s self-report under ablation ($0.880 \rightarrow 0.709$). And the pre-attempt read, taken where
294 the activation is byte-identical across a question’s attempts, cannot recover the post-attempt
295 direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would
296 predict it could. A linear probe at one token also undercounts nonlinear or multi-token
297 signal, so the reported numbers are a floor.

298 D Per-cell numbers and specificity controls

Table 2: Post-attempt correctness discrimination (AUROC). Left: stated confidence vs. the probe in representative cells; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Section B). Right: Mistral specificity controls. “pre” is the last prompt token before the attempt; “post” is after the attempt; “resid.” removes the pre score; “post→pre” applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

model × domain	stated	probe	Mistral	pre	post	resid.	post→pre
Gemma-4 × MATH	0.57	0.78	MMLU-Pro	0.568	0.754	0.751	0.490
Qwen3.6 × MMLU-Pro	0.67	0.84	MATH	0.792	0.834	0.701	0.624
GLM × MMLU-Pro	0.67	0.85	GPQA	0.593	0.572	0.544	0.479
Mistral × MATH	0.79	0.83	GeoGen	0.770	0.676	0.505	0.648